MIL-P-14737 (MU)
4 April 1968
SUPERSEDING
PA-PD-46 (Rev. 1.)
1 June 1956

MILITARY SPECIFICATION
PROPELLANT, M7

1. SCOPE

1.1 This specification covers standard M7 rocket propellant for use in rockets and modified M7 rocket propellant for use in the LAW program.

1.2 Classification

TYPE I - Standard M7 rocket propellant
TYPE II - Modified M7 rocket propellant

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-217 - Potassium Perchlorate
MIL-N-244 - Nitrocellulose
MIL-N-246 - Nitroglycerin
MIL-E-255 - Ethyl Centralite (Carbamite)
MIL-C-306 - Carbon Black, Dry, (For Use in Explosives)
MIL-E-463 - Alcohol, Ethyl (For Ordnance Use)
MIL-A-489 - Acetone (For Ordnance Use)
MIL-M-542 - Magnesium Stearate (For Use in Ammunition)
MIL-M-12036 - Magnesium Oxide, Light, Technical

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-286 - Propellants, Solid: Sampling, Inspection and Testing
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

FSC: 1376

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Publication.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations (1949 Edition and Revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 71-90 (latest revision)".)

3. REQUIREMENTS

3.1 Materials.-The materials shall comply with the requirement of the applicable specification as follows:

Material	Specification
Potassium Perchlorate	MIL-P-217, Grades A and B, all types, to be ground to an average particle diameter of 6 to 9 microns and blended with 0.5% magnesium oxide 0.5% magnesium stearate, of like particle diameter.
Nitrocellulose	MIL-N-244, Grade C, Type I
Nitroglycerin	MIL-N-246, Type I
Ethyl Centralite	MIL-E-255
Carbon Black	MIL-C-306
Alcohol, Ethyl	MIL-E-463, Grade 2
Acetone	MIL-A-489
Magnesium Stearate	MIL-M-542
Magnesium Oxide	MIL-M-12036

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3.2 Composition.-The composition shall conform to the requirements of Table I when tested as specified in Section 4.2.3.1.

TABLE I

Standard Propellant Material (Type I)	Percent
Nitrocellulose	54.60 Nominal
Nitroglycerin	35.50 Nominal
Potassium Perchlorate	8.05 Maximum
Carbon Black	1.20 Nominal
Ethyl Centralite	0.80 Minimum

Modified Propellant Material (Type II)	Percent
Nitrocellulose	58.70 Nominal
Nitroglycerin	32.00 Nominal
Potassium Perchlorate	8.05 Maximum
Ethyl Centralite	0.80 Minimum
Carbon Black	0.60 Nominal

3.3 Total volatiles.-The total volatiles content of the finished propellant, standard and modified, shall not exceed 0.80 percent when determined as specified in 4.2.3.2.

3.4 120.0°C Heat Test.-Neither the standard nor modified propellant shall cause complete fading of methyl violet paper to salmon pink in less than 40 minutes when determined as specified in 4.2.3.3.

3.5 Heat of explosion.-The heat of explosion of the finished propellant, standard and modified, shall not exceed a maximum of 1290 calories per gram nor fall below a minimum of 1260 calories per gram when determined as specified in 4.2.3.4.

3.6 Workmanship.-The best commercial practices shall be used in formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes during and subsequent to its manufacture. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified. The propellant furnished under this specification shall be free from pin holes, blisters, sub-surface air occlusion, cracks or surface imperfections exceeding 1/16 inch in length.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or military specification, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet AMC Form 1047 (OO Form 1204) containing the applicable data (see 6.3).

4.1.2 Certificate to be submitted.-The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection provisions.

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which a representative sample (see 4.2.2.1) is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more batches or propellant produced by one manufacturer, in accordance with the same specification, and same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168. The lot size of propellant deliverable under the contract shall not be less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period.

4.2.2 Testing

4.2.2.1 Sampling.-Two sixty (60) gram samples shall be selected at random from each lot of propellant. The samples shall each be checked for the requirements specified in paragraphs 3.2 through 3.5. The nominal percentages for nitrocellulose, nitroglycerin and carbon black stipulated in Table I of 3.2 are not requirements. For the first three initial production lots (see 6.4), the sample size shall be increased to 80 grams to allow for three additional heat of explosion tests.

4.2.2.2 Rejection.-If any individual sample fails to meet the requirements specified in paragraphs 3.2 through 3.5, the lot of propellant shall be rejected.

4.2.3 Test methods and procedures

4.2.3.1 Composition

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4.2.3.1.1 Potassium Perchlorate.-Major Defect, Code No. 01001 (see 6.2).-The potassium perchlorate content shall be determined in accordance with Standard MIL-STD-286, Method 310.4.

4.2.3.1.2 Ethyl Centralite, Major Defect, Code No. 02001.-The ethyl centralite content shall be determined in accordance with Standard MIL-STD-286, Method 202.2.

4.2.3.2 Total Volatiles, Major Defect, Code No. 03001.-The total volatile content shall be determined in accordance with Method 103.3 of Standard MIL-STD-286.

4.2.3.3 120.0 Degree C. heat test, Major Defect, Code No. 04001.-The 120.0 degree heat test shall be conducted as specified in Method 404.1 of MIL-STD-286.

4.2.3.4 Heat of explosion, Major Defect, Code No. 05001.-The heat of explosion shall be determined in accordance with Method 802.1 of Standard MIL-STD-286.

5. PREPARATION FOR DELIVERY

5.1 Packing.-The propellant shall be packed in accordance with ICC Regulations and the best commercial practice.

5.2 Marking.-The containers shall be marked in accordance with ICC Regulations and MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1).
c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Submission of AMC Form 1047.-Information contained in propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents are adhered to (see 4.1.1).

6.4 Initial production lots.-The initial production lots represent the first three consecutive lots which have been produced and accepted.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 1376-A-011

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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